CLEW

AGENTIC DEVELOPMENT · 2026

The model is the engine. The harness is the car.

What makes a good agentic coding harness, and how clew keeps it coherent.

SECTION 01 / 05

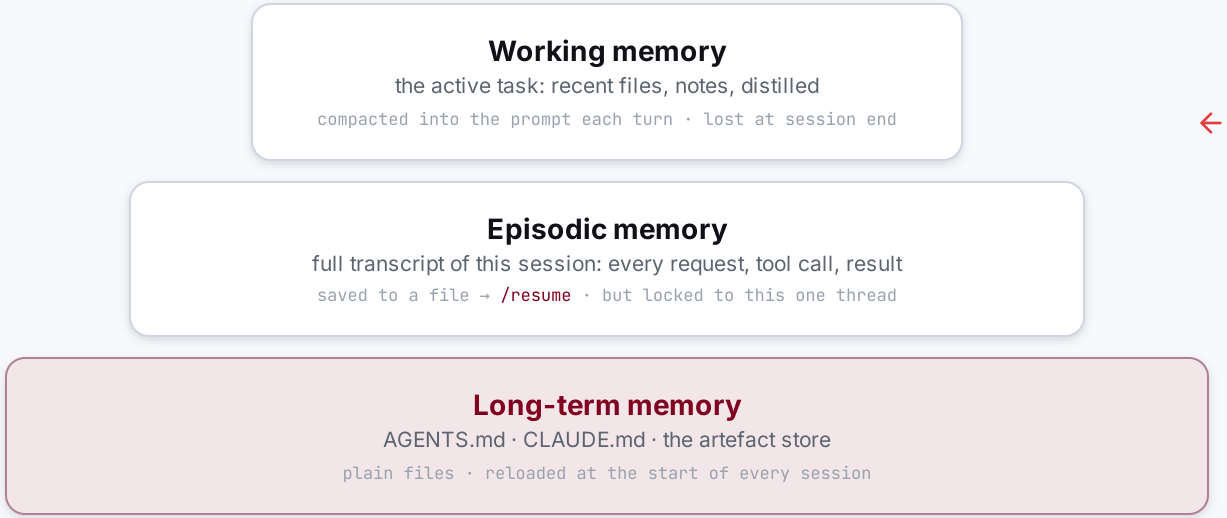
The Problem

Agents can't remember between sessions, and every obvious fix for it fails.

01

THE GAP

Most harnesses only have the shallowest memory, yet memory is the moat.



← most stop here

the moat

2 of 3 session-bound

Working memory is distilled, then discarded at session end; episodic is a resumable transcript but locked to its own thread. Only long-term (plain files) crosses into the next session.

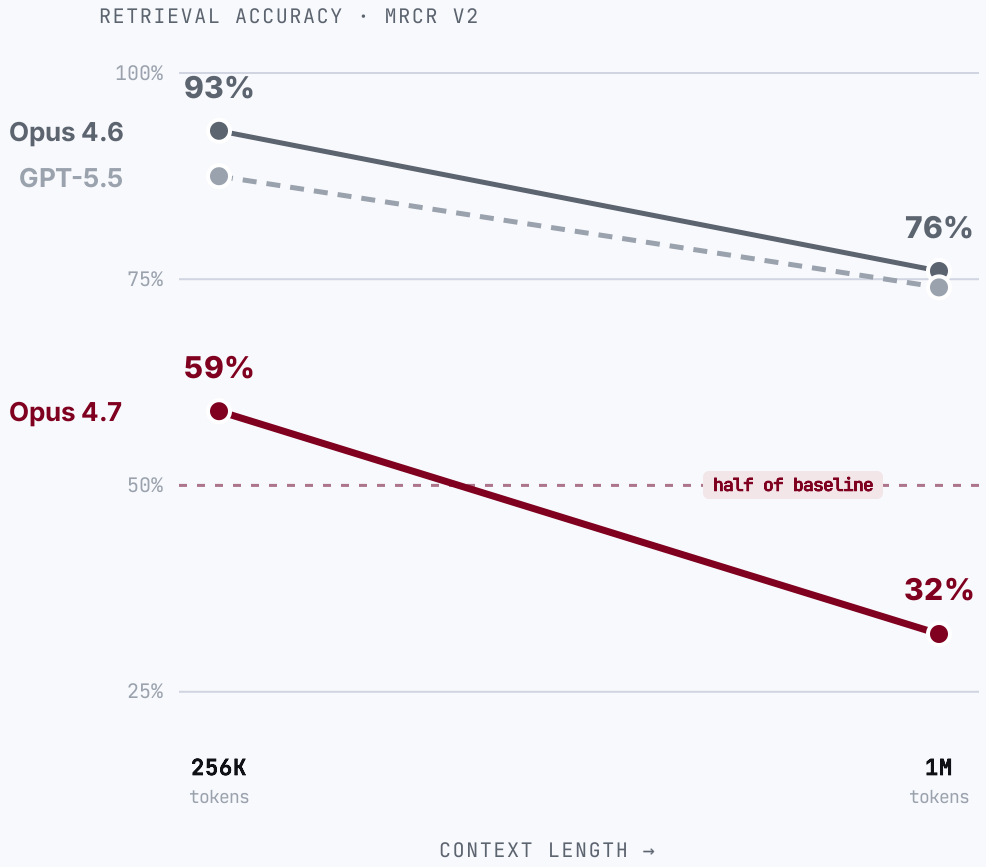
0 tokens / fetch

And it's a local file read, no MCP API roundtrip, no token cost, reloaded at the start of every session.

THE OBVIOUS OBJECTION *"A frontier window holds a million tokens now, so why not just let the window be the memory?"* ANSWERED NEXT →

No. A bigger context window is not bigger memory.

Bury one fact in a million-token prompt and ask for it back: the frontier model returns it barely **1 in 3** times. The window **forgets** as you fill it.



The advertised window is **not** the usable window.

59% → 32%

Claude Opus 4.7 loses ~half its multi-needle retrieval from 256K to 1M tokens. Opus 4.6 holds better (93% → 76%).

Codex too

the GPT-5.x line behind Codex CLI drops ~87% → 74% at long context, a proxy; no per-length figure is published for the Codex-tuned model.

11 of 13

models fell below half their accuracy by 32K tokens. Context rot isn't new (NoLiMa, 2025).

Anthropic · Claude Opus 4.7 System Card · MRCR v2 (vendor-reported) (2026) — anthropic.com | Anthropic · Claude Opus 4.6 System Card · MRCR v2 (vendor-reported) (2026) — anthropi…

Writing it down isn't memory either.

OpenAI shipped a 1M-line product with **zero** hand-written code, then found the obvious fix, one big context file, fails.

1M lines of code | **0** hand-written | **1,500** PRs merged

"One big `AGENTS.md`" failed four ways

- **Crowds out the task.** Context is scarce: the giant file pushes out the code and the docs.
- **Too much guidance becomes none.** When everything is "important," nothing is.
- **It rots instantly.** A graveyard of stale rules: the agent can't tell what's still true.
- **It can't be verified.** Drift is inevitable without mechanical checks.

If it's not in-context, it doesn't exist

Anything in Slack, Google Docs, or someone's head is invisible to the agent.

OpenAI's fix

A **~100-line table of contents** pointing into a **structured `docs/` tree**: a map, not an encyclopedia.

THE LEVER · REALITY CHECK

2025 was the “Year of the Agent.” It didn't land.

The hype was model-led, but in production, the model alone fell short, expensively.

95%

of enterprise GenAI pilots showed no measurable P&L impact

MIT studied 300 deployments + 350 leaders. The blocker wasn't model IQ; it was integrating AI into real workflows.

~59%

is the best frontier model's score on realistic enterprise tasks

Scale's SWE-bench Pro: 1,865 real tasks across 41 professional repos (2026). The same models score 80–95% on the benchmark everyone quotes; the gap is the reality.

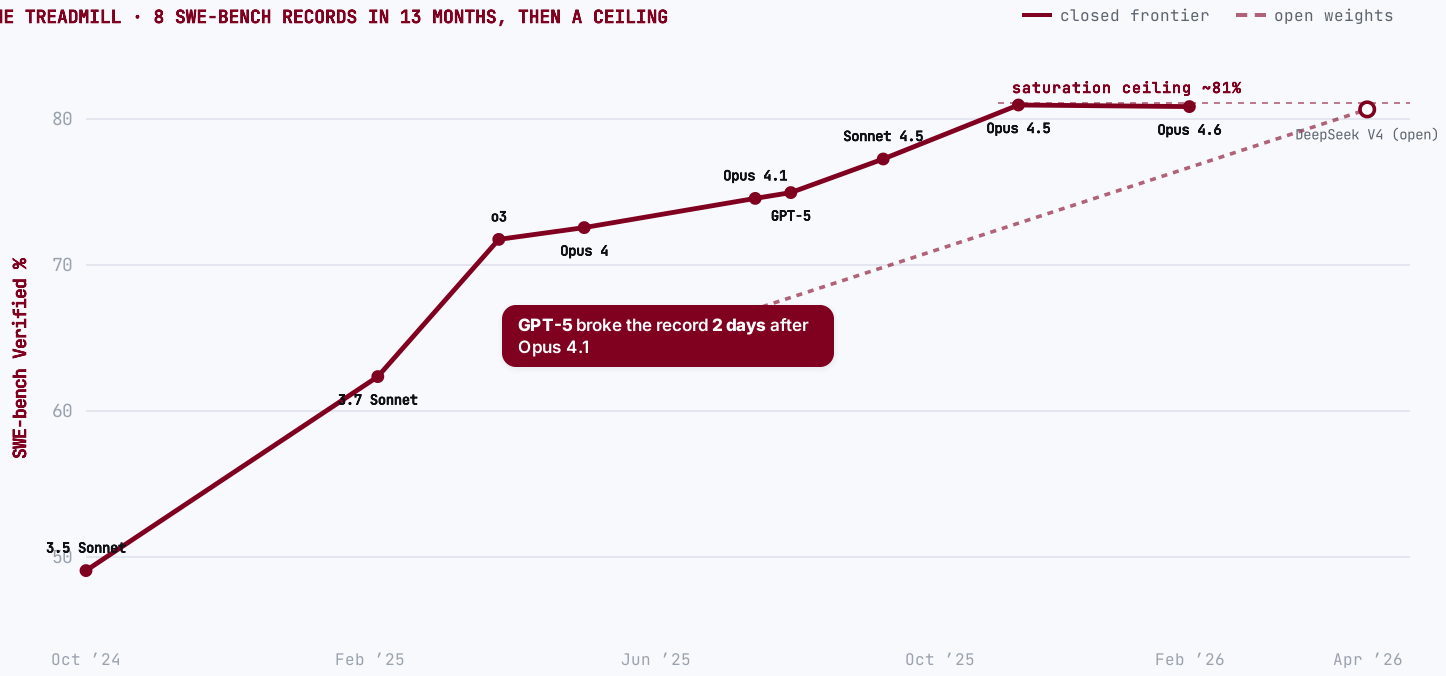
The missing variable wasn't intelligence; it was the **harness around the model**.

MIT Project NANDA · The GenAI Divide: State of AI in Business 2025 (95% of pilots: no measurable P&L impact) · via Fortune (2025) — fortune.com | Scale AI (SEAL) · SWE-bench Pro:…

Models leapfrog every benchmark, then commoditize.

A new frontier model tops every benchmark… until the next does, **days** later. Then it saturates ~81%, open weights reach **parity**, and the benchmark itself gets retired.

THE TREADMILL · 8 SWE-BENCH RECORDS IN 13 MONTHS, THEN A CEILING



Open weights, now at parity

≈ 80.6% = the frontier

DeepSeek V4-Pro matches the closed SOTA on SWE-bench Verified (Apr '26). Composite gap now ~**4 months / 8 ECI pts** (Epoch AI, May '26).

The benchmark itself broke

59% of hard tests flawed

OpenAI **retired SWE-bench Verified** (Feb '26): frontier models had trained on the answers; it's saturated. → SWE-bench Pro.

Inference is collapsing

≈ 280× cheaper / 18 mo

GPT-3.5-level tokens: $20.00 → $0.07 per M. ~10× every year: "LLMflation."

> ***"The models are getting commoditized."***
>
> Satya Nadella · Microsoft CEO · BG2 Pod, Dec 2024

The model is a commodity, not a moat.

Whatever tops the leaderboard this week is matched, undercut and open-weighted within months, and the benchmark itself gets retired.

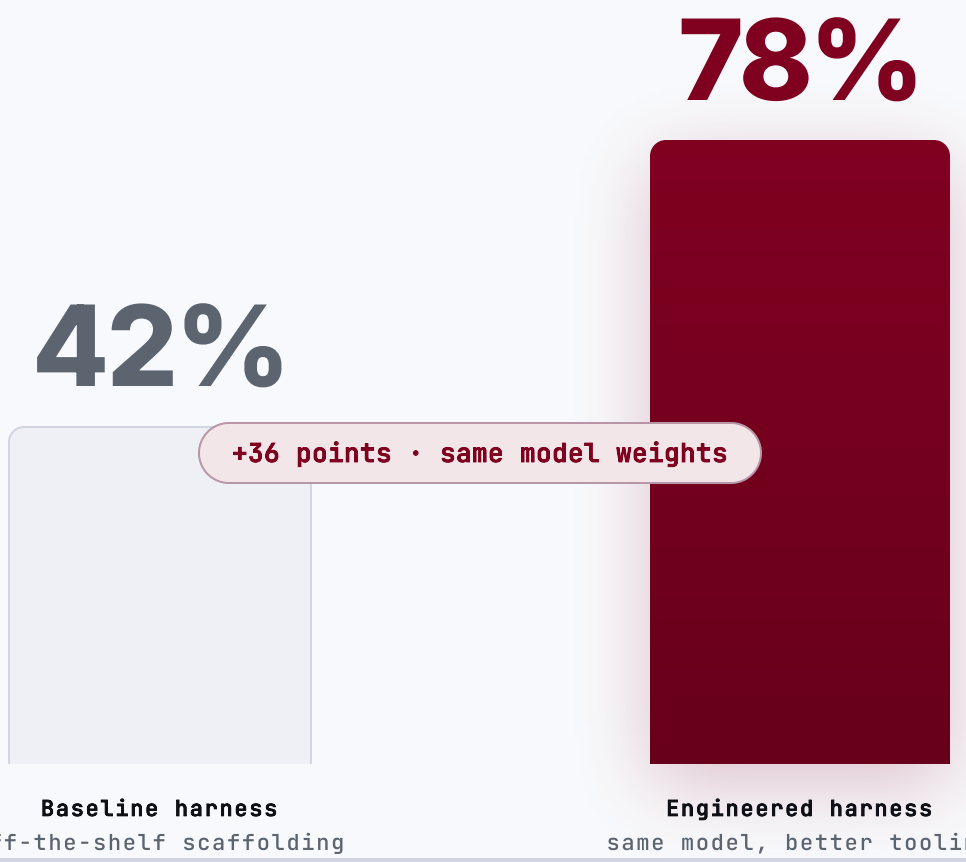
SECTION 02 / 05

The Agentic Harness

Why the tooling around the model, not the model, is the performance lever.

02

Hold the model fixed, swap the harness: 42% → 78%.

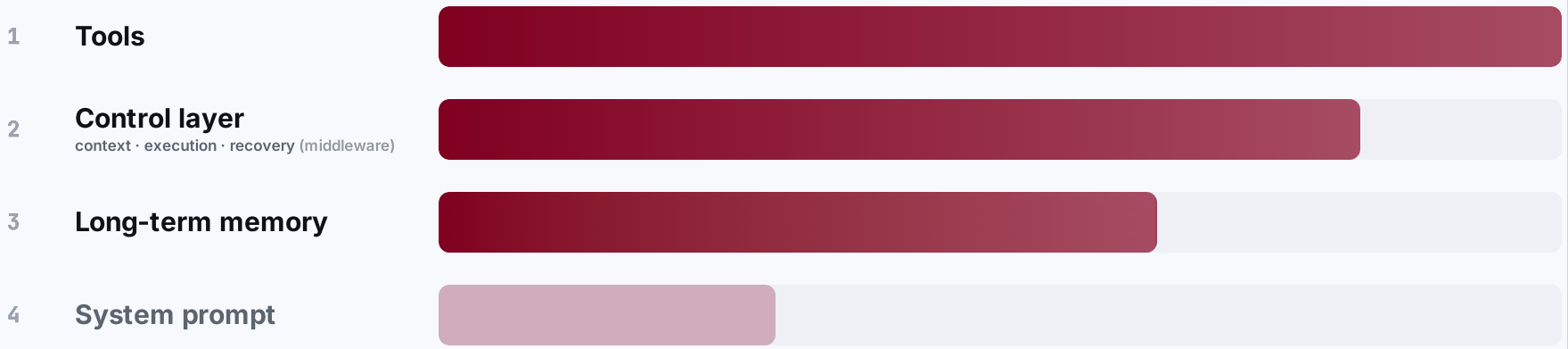


Claude Opus 4.5 · HAL CORE-Bench · the only thing that changed was the harness

THE LEVER · WHERE IT LIVES

Performance lives in the tooling around the model, not the model.

Ablation studies rank what actually moves agent performance. Model prompt-tuning sits **last**.



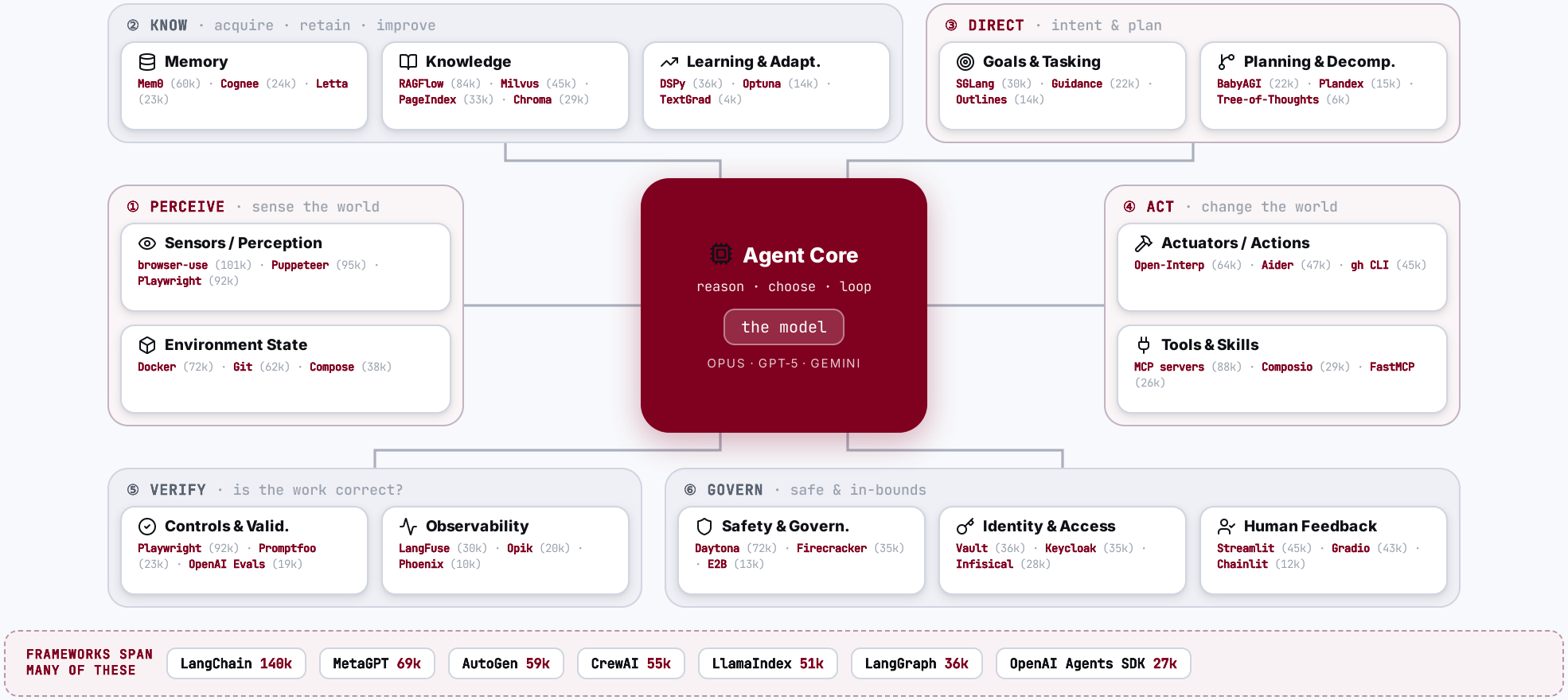
THE INVERSION

Most teams pour effort into #4, the prompt. The gains sit in #1–3.

THE MODEL

Table stakes once it is capable. The harness is the differentiator.

The agent core, wrapped in *14 capabilities*, and who leads each.



THE LEVER · THE HONEST OBJECTION

Do you really think software engineering is dead?

That map was **50 tools** wrapping a single model: each its own system, all to be wired together and kept alive.

50

tools around one model

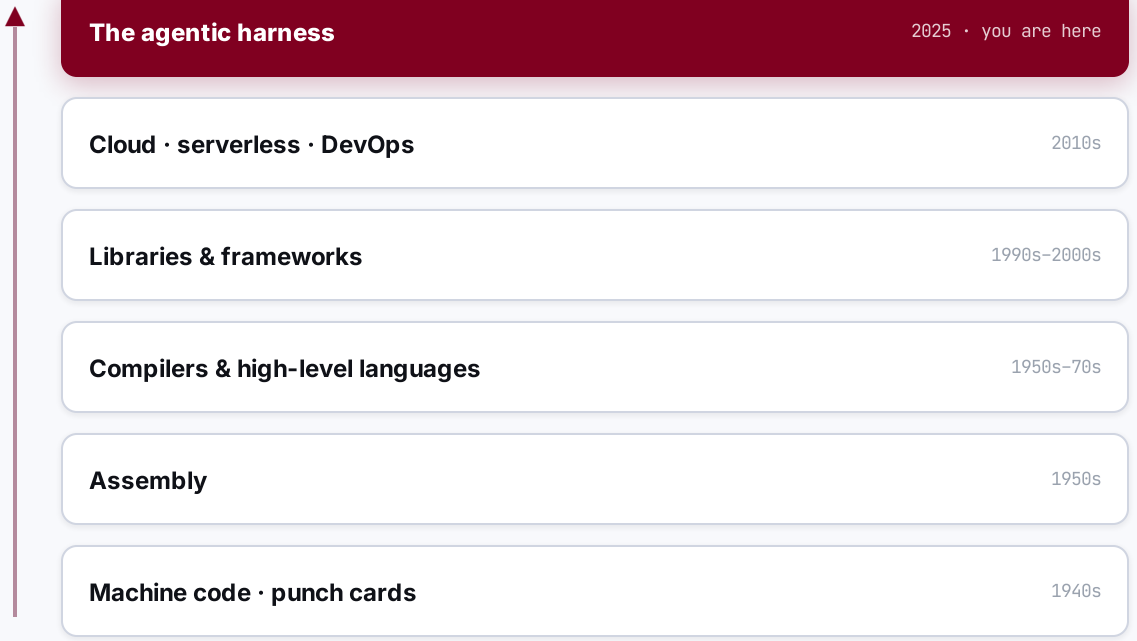
and only the top few per capability; 80+ ship in the wild

Each tool is its own system
its own API, config, failure modes and learning curve

Orchestration is the hard part
wiring 50 moving parts into one reliable, debuggable loop

Maintenance never stops
versions drift, tools break, the harness needs constant upkeep

THE LADDER OF ABSTRACTION · every rung was once "the end of programming"



No. We've just climbed another rung on the ladder of abstraction.
Software engineering didn't disappear; building and maintaining this harness **is** the engineering now.

Internal research · Agentic Harness: Capabilities map (15 functional domains, 80+ real shipping tools) (2026)

BUILDING WITH AGENTS · IN PRACTICE

Dev Tooling

The editor, model, agent-CLI and IDE you build with, open-source or proprietary, and how engineers climb from vibe-coding to real agentic engineering. The harness and the method underneath stay the same.

THE BUILD WORKFLOW · YOUR SETUP

Open-source or proprietary: your setup, your call.

Three tiers: bring a model, compose a custom harness, or run a full IDE, each available **fully open** or **fully proprietary**. What doesn't change is the method underneath.

	OPEN-SOURCE · self-hostable	PROPRIETARY · hosted
Full IDE ALL-IN-ONE	VS Code ⋆187k · Zed ⋆86k · Continue ⋆35k · Tabby ⋆34k Zed standalone · VS Code + ext	Cursor · Windsurf · Antigravity · Kiro closed-source · bundled
Custom harness EDITOR + AGENT	opencode ⋆180k · Cline ⋆64k · Goose ⋆50k · Aider ⋆47k · Crush ⋆26k open + model-agnostic	Claude Code ⋆135k · Codex ⋆94k · Copilot · Antigravity CLI · Amp proprietary · vendor-hosted
Inference & routing OPTIONAL	Ollama ⋆175k · llama.cpp ⋆118k · vLLM ⋆85k · LiteLLM ⋆52k self-host · open	OpenRouter · Groq · Hugging Face · Bedrock · Vertex AI · Azure OpenAI managed · cloud gateway
Bring your own model THE MODEL	DeepSeek · Llama · Qwen · gpt-oss · Mistral open weights · self-hostable	Claude · GPT-5 · Gemini 3 · Grok frontier · hosted

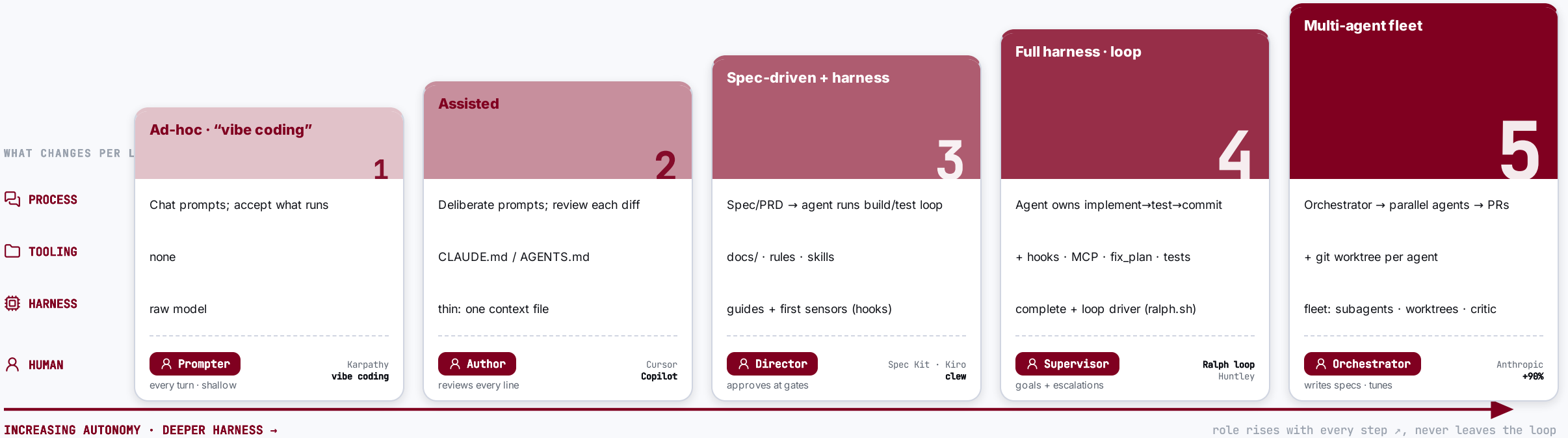
The method THE CONSTANT, EITHER SIDE

`spec` → `plan` → `grill-me` → `peer-review` → `loop` · artefacts · rules · skills · the **metamodel**

GitHub · Repository stargazers: live API snapshot (tool popularity) (2026) — api.github.com | Google · Antigravity: agentic development platform (IDE + CLI), launched 2025-11-18 …

From “vibe coding” to *agentic engineering*: five levels.

Autonomy climbs and the harness thickens, but the human moves **up** the stack (prompter → orchestrator), never out of it.



⚠ **Even L5 is *supervised* autonomy, not hands-off.** METR: experienced devs were 19% *slower* on mature codebases; SWE-bench Pro tops out ~23%; Ralph’s author: “no way I’d use it on an existing codebase.” The guardrail against Thoughtworks’ **“cognitive debt”** is keeping humans on standards, architecture & final sign-off.

SECTION 03 / 05

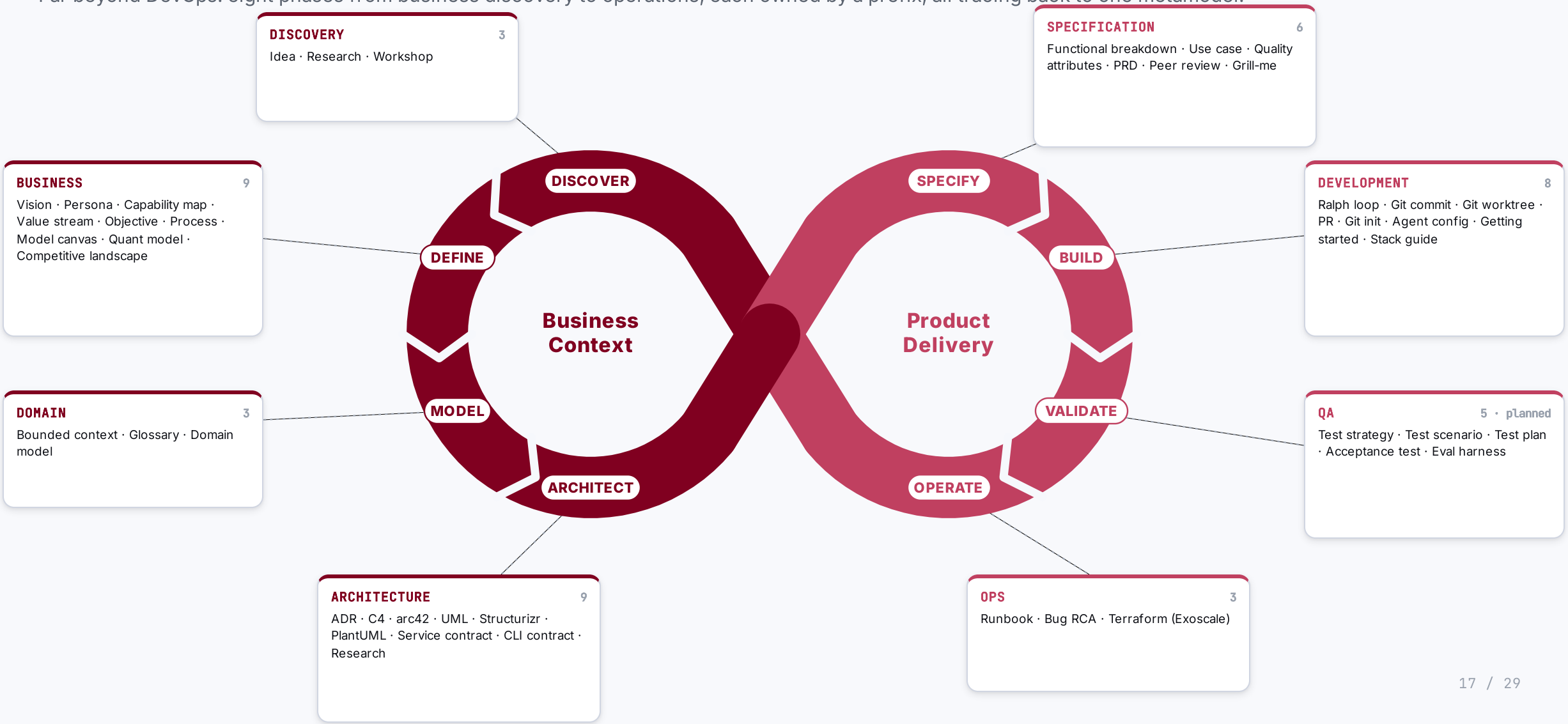
The Metamodel

How an agent armed with skills and rules turns business context into deployable artefacts, every step traceable.

03

Skills & rules wrap the whole lifecycle, from discovery to operations.

Far beyond DevOps: eight phases from business discovery to operations, each owned by a prefix, all tracing back to one metamodel.



Discover & Define · make the context real.

WHAT THE AGENTIC ENGINEER DOES HERE

- **Brainstorm & capture**
  ideas, hypotheses
- **Research & scan**
  market, competitors
- **Interview & workshop**
  with the business
- **Model the business**
  vision → objectives
- **Define the domain**
  ubiquitous language

Business
9 SKILLS

- **Vision** — north star
- **Persona** — who we serve
- **Model canvas** — value model
- **Capability map** — what it can do
- **Competitive landscape** — market position
- **Value stream** — how value flows
- **Objective** — goals & KRs
- **Process** — operations
- **Quant model** — unit economics

Discovery
3 SKILLS

- **Idea** — capture & graduate
- **Research** — 1:1 interviews
- **Workshop** — group facilitation

Domain
3 SKILLS

- **Bounded context** — boundaries
- **Glossary** — ubiquitous language
- **Domain model** — aggregates & events

Architect the solution · turn context into a solution.

WHAT THE AGENTIC ENGINEER DOES HERE

- **Decide architecture** — ADRs, C4, arc42
- **Define contracts** — service, CLI
- **Break down function** — FBS
- **Specify & qualify** — use cases, quality
- **Write & review PRDs** — one per epic

Architecture
9 SKILLS

- **ADR** — decisions
- **Research** — arch notes
- **Service contract** — API surface
- **CLI contract** — command surface
- **Structurizr** — C4 workspace
- **C4** — context → component
- **PlantUML** — UML pipeline
- **UML** — sequence/class/state
- **arc42** — arch narrative

Specification
6 SKILLS

- **Functional breakdown** — feature tree
- **Use case** — actor scenarios
- **Quality attributes** — ISO 25010
- **PRD** — one per epic
- **Peer review** — PRD/plan review
- **Grill-me** — adversarial review

UX
1 SKILL

- **Design system** — tokens & brand

Execute & Deliver · plan, build, test, ship.

WHAT THE AGENTIC ENGINEER DOES HERE

- **Sequence the roadmap**
  epics, phases
- **Plan increments**
  atomic steps
- **Build & validate**
  tests, checks
- **Deploy**
  infra, release
- **Operate & fix**
  runbooks, RCA

Specification
2 SKILLS

- **Delivery roadmap**
  epics & phases
- **Implementation plan**
  atomic increments

QA
5 · PLANNED

- **Test strategy**
  pyramid & coverage
- **Test scenario**
  from use cases
- **Test plan**
  per release
- **Acceptance test**
  Given–When–Then
- **Eval harness**
  agent evals

Development
8 SKILLS

- **Agent config**
  CLAUDE.md & rules
- **Getting started**
  onboarding
- **Stack guide**
  tech patterns
- **Git init**
  repo setup
- **Git commit**
  commits
- **Git worktree**
  worktrees
- **PR**
  pull requests
- **Ralph loop**
  autonomous loop

Ops
3 SKILLS

- **Runbook**
  operations
- **Bug RCA**
  incident analysis
- **Terraform**
  Exoscale infra

fed by ← Design

SECTION 04 / 05

The Build Workflow

From a PRD to merged code: how one engineer drives agents through alignment and verification gates, then lets them run autonomously.

04

THE BUILD WORKFLOW · END TO END

From PRD to merged code: two fresh-session gates, then autonomy.

Business context and discovery are done; the roadmap gives one PRD per epic. The loop below runs per PRD.

1 Spec ready

ALREADY DONE

- Delivery roadmap → **one PRD per epic**
- Each PRD states what must be true

PRD in hand

2 Align & plan

FRESH SESSION

- `agent-grill-me`: agent grills you ~10 Q on impl & architecture
- you answer → shared mental model
- agent writes the **implementation plan**

Plan drafted

3 Verify the plan

FRESH SESSION

- `agent-peer-review`: findings ranked critical → low
- you resolve the ones needing guidance → amend
- `ralph audit`: plan is autonomously runnable

Audit green

4 Execute & ship

SEPARATE CLI

- `agent-ralph-loop ralph.sh`: fresh agent per increment
- implement → test gate → commit, until done
- run CI on the finished work

Merged

Two **fresh-session** checks (grill-me before, peer-review after) keep me and the agent aligned; the loop only runs once the plan is proven autonomous.

PRD says what must be true; the plan says how, safely.

Two artefacts, one chain: the PRD scopes the epic, the implementation plan breaks it into runnable increments.

BUILD BY FEATURE · ONE PER EPIC

PRD

"What must be true, and why."

- **Problem, goals & non-goals**: the scope boundary
- **User stories** grounded in personas (P-NN)
- **Acceptance criteria** + success metrics
- **§0 traceability**: the epic and its dependencies

E-NN · P-NN · C-N.M.FXX · QA-XXNN · UC-NN

1 PRD
↓
1 plan

acceptance criteria
become
the plan's test gates

increments
deliver
the PRD scope

ATOMIC INCREMENTS · ONE PER PRD

Implementation plan

"How we get there, reversibly."

- **Ordered increments** (Inc-1 … Inc-N)
- each **small · testable · reversible**
- scope + primary files per increment
- **test gate** + exit criteria = done

Plan-NNNN · Inc-N · Status: pending → done

Two fresh agents pressure-test the work, before and after the plan.

Grill-me pulls the thinking out of my head; peer-review checks the document. Both run in a clean session.

ALIGNMENT · BEFORE THE PLAN

agent-grill-me

Live Socratic session on the PRD, one question at a time.

- Agent asks ~10 focused Q on **implementation & architecture**
- Vocabulary checked against the **domain glossary**
- Decisions crystallised into **ADRs** while context is live
- Surfaces the **thinking gaps** only questioning reveals

Output: **shared mental model** → the agent then writes the implementation plan.

VERIFICATION · AFTER THE PLAN

agent-peer-review

Static, independent scan of the finished plan.

- Reads the plan end-to-end, builds a requirement map
- Findings ranked with concrete remediation:

critical major normal low

- I resolve the ones needing guidance → **amend the plan**

Output: **validated plan** → ready for the autonomy audit.

Peer-review finds the **mechanical** gaps automatically; grill-me surfaces the **thinking** gaps under questioning.

The Ralph loop: a fresh agent per increment, until the plan is done.

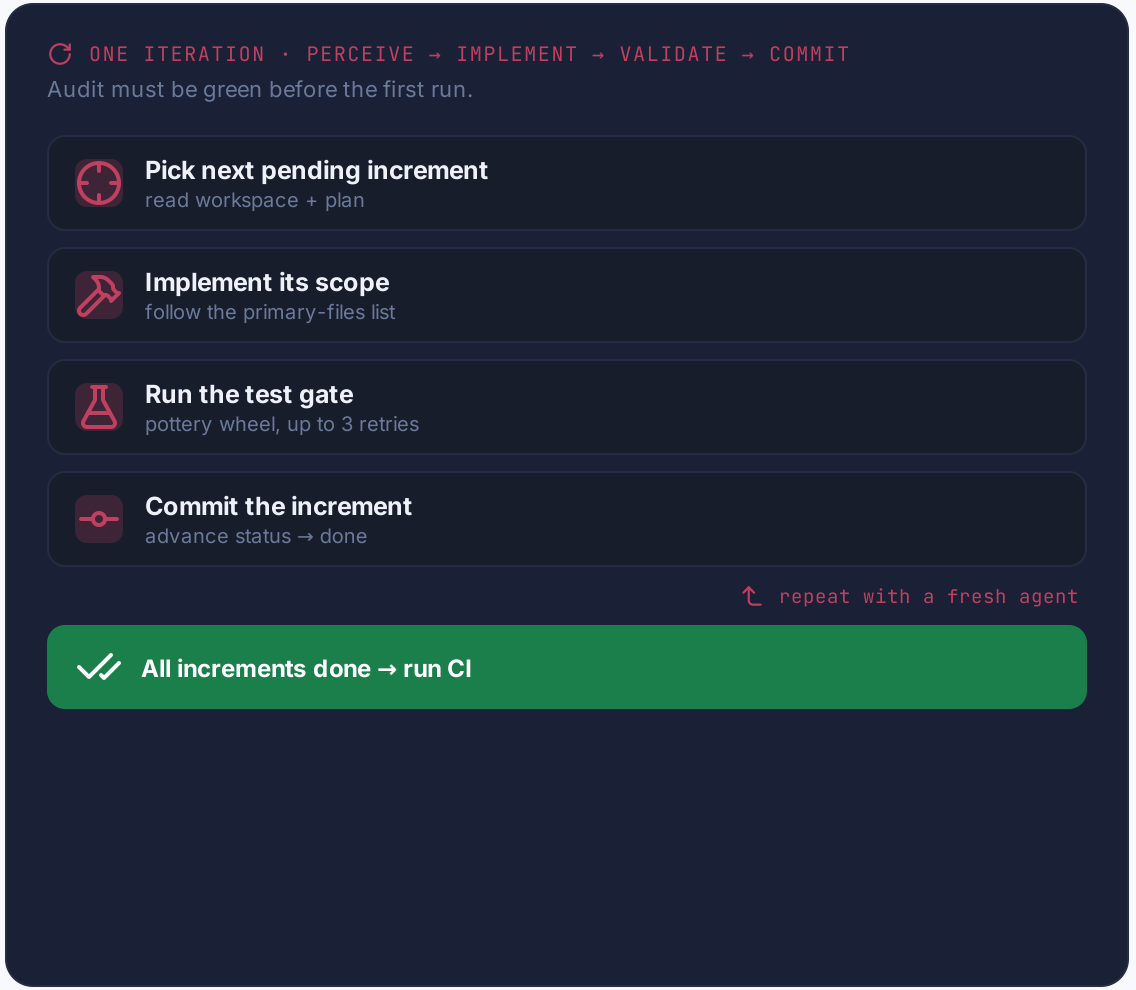
WHAT THE LOOP IS

A bash script (**ralph.sh**) drives the plan to completion, one increment per iteration.

Fresh agent each iteration → clean context, no drift or context rot.

Loops while increments are **pending**; stops when every one is **Status: done**.

```
ralph.sh <workspace> \
  --max-iterations 50 --with-prd --with-push
# spawns: claude --print < iteration-prompt
```



BEFORE YOU GO

Key takeaways

Ten commandments for the developer, ten for the company. What to actually do on Monday.

THE TAKEAWAY · FOR THE DEVELOPER

Ten commandments for the developer.

The agent is a power tool. The judgement, the architecture and the guardrails are still yours.

1. **Build your own harness, but share what works.**
   No one tells you which IDE or agent to pick; pass the patterns and best practices back to the team.
2. **Treat your agent like a smart junior.**
   Give it tools, steer it, correct it.
3. **Keep a fresh context.**
   Start clean; don’t let the window rot.
4. **Agents magnify bad architecture as fast as good.**
   So build the guardrails.
5. **The tooling is hackable.**
   It was built to be bent; adapt it to your workflow.
6. **Start minimal; earn every tool and framework.**
   Hit a wall before you add a tool; run the agent loop bare before you adopt an agentic framework.
7. **Fast to write is not a reason to write.**
   Coding was never the bottleneck; thinking was.
8. **Trust, but verify.**
   Read the diff; every line is yours to defend.
9. **Learn by playing.**
   Build hobby projects, not experiments on company goals.
10. **Refactoring got cheap. Architecture didn’t get optional.**
    Cheap to change is not the same as fine to wing it.

⚓ **Engineer the harness, not just the prompt. The craft moved up a rung; it didn’t disappear.**

THE TAKEAWAY · FOR THE COMPANY

Ten commandments for the company.

Building with agents is an organisational capability, not a gadget. Set the conditions.

1. **Enable the tooling.**
   Give every dev agents, sandboxes and access; don't gatekeep the lever.

2. **Standardize the workflow and artefacts, not the tool.**
   Mandate the method; let people pick their own harness.

3. **AI skips none of your real problems.**
   Quality, process, planning, architecture are still there; you just hit the wall faster, and it's bigger.

4. **Claim AI sovereignty now.**
   Don't turn every model and tool into a US dependency.

5. **Default to open source.**
   OSS always ends up the default; don't repeat the Microsoft Office lock-in.

6. **Govern the data before the agent sees it.**
   Decide what company data an agent may ingest; first, not after.

7. **Budget for maintenance.**
   The harness is a living system, not a one-off.

8. **Pay down cognitive debt.**
   Don't let agents quietly erode the team's understanding.

9. **Measure what must be true, not velocity.**
   Track traceability and correctness, not tickets closed.

10. **The method is the moat, not the model.**
    Invest in the harness and the memory; models are commodities.

⚓ The model is a commodity. Your edge is the **method and memory** you build around it.

THE TAKEAWAY

Engineer the harness, not just the prompt.

Start with the highest-leverage capability: memory that doesn't rot.

- Not a PM tool
- Not a wiki
- Not a SaaS
- The repo is the source of truth

Git for product architecture: manage what must be true, not what's in progress.